

■ TEAM OVERFIT GOBLINS

ÉquiAlgo: fair student financing

A challenge by IVADO · Engineering and Computer Science Hackathon 2026



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Same model. Different outcomes.

A Quebec financial institution scores scholarship and student-loan applications with a model that is 88% accurate.

It learned from a committee's past decisions. Those decisions treated two regions differently, and the model copies them.

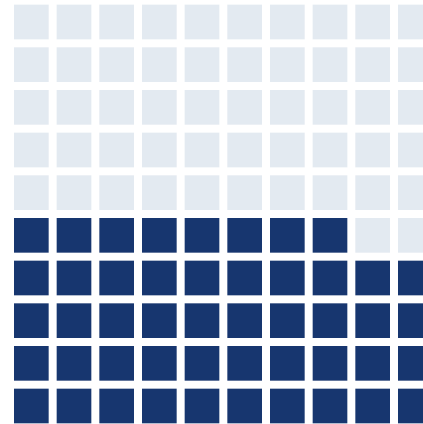
Δ 21 points between Centre and Remote

■ CENTRE

Montréal
Capitale-Nationale

48.4%

granted a scholarship

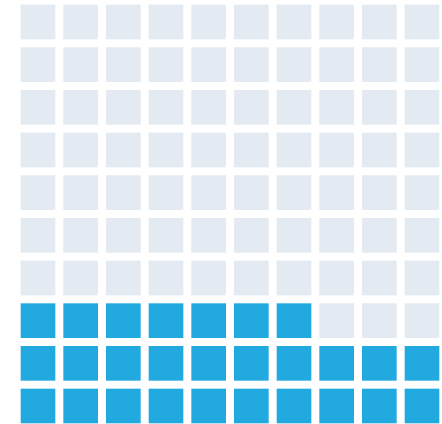


■ REMOTE

Bas-Saint-Laurent
Côte-Nord
Gaspésie-Îles-de-la-Madeleine

27.3%

granted a scholarship



■ OUR APPROACH

From prediction to evidence.

Four steps: make the bias visible, explain it, fix it, and keep watching.

1

Measure the bias

VISIBLE · MEASURABLE

Grant rates by R-score band, and an equal-opportunity metric, for the committee and the baseline model.

2

Find the proxies

EXPLAINABLE

Which columns still reveal region once the region column is removed? We tested 127 feature combinations.

3

Build the fix

ACTIONABLE

A qualified score, a fairness step per region and a fixed grant budget. Results on a Pareto plot.

4

Keep watching

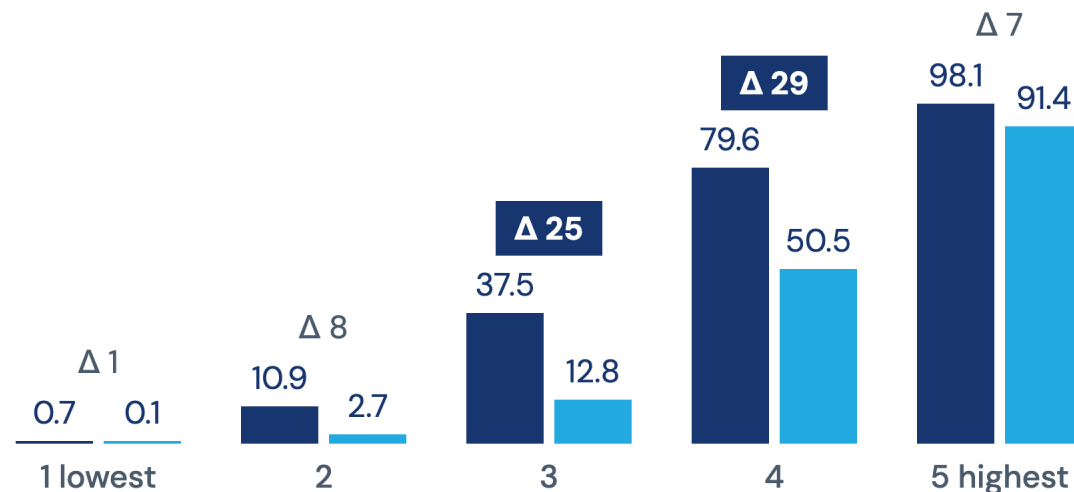
RESPONSIBLE

A monitoring plan for production: gaps tracked by region, alerts, and humans reviewing denials.

Same grades. Different outcomes.

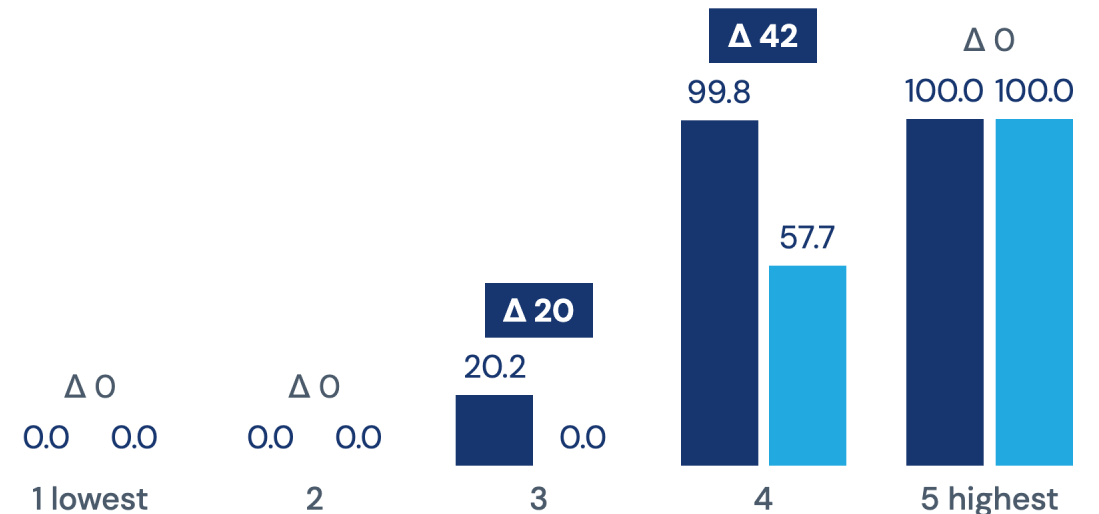
Grant rate by R-score band. The baseline model copies the committee: at the same grades, Remote applicants still get fewer grants.

Past committee decisions



Largest gap: 29 points, band 4

Baseline model predictions



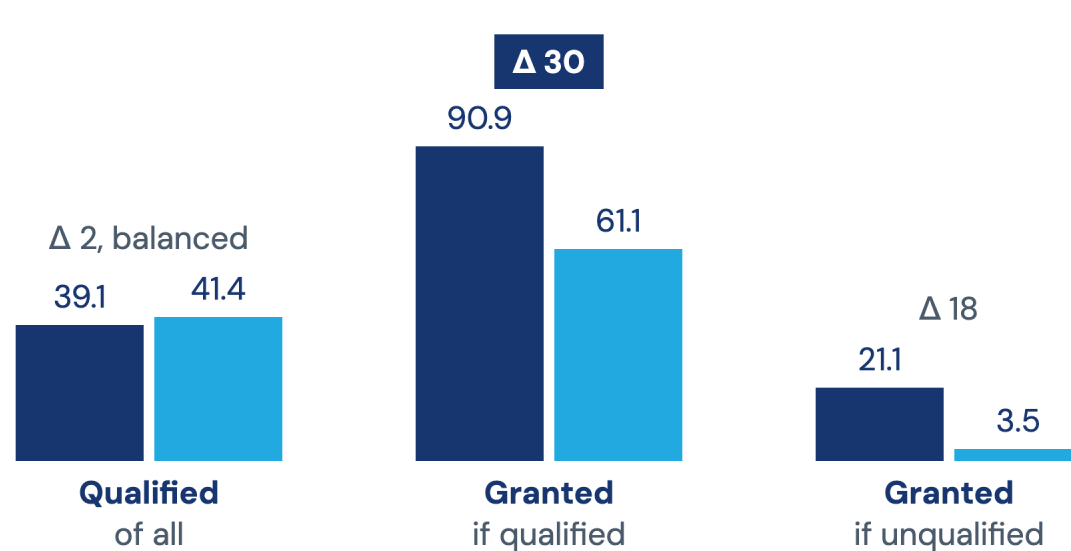
Largest gap: 42 points, band 4

Fairness metrics

Qualified score = $(3 \times \text{R score} + 0.5 \times \text{income need} + 0.5 \times \text{hours burden}) / 4$

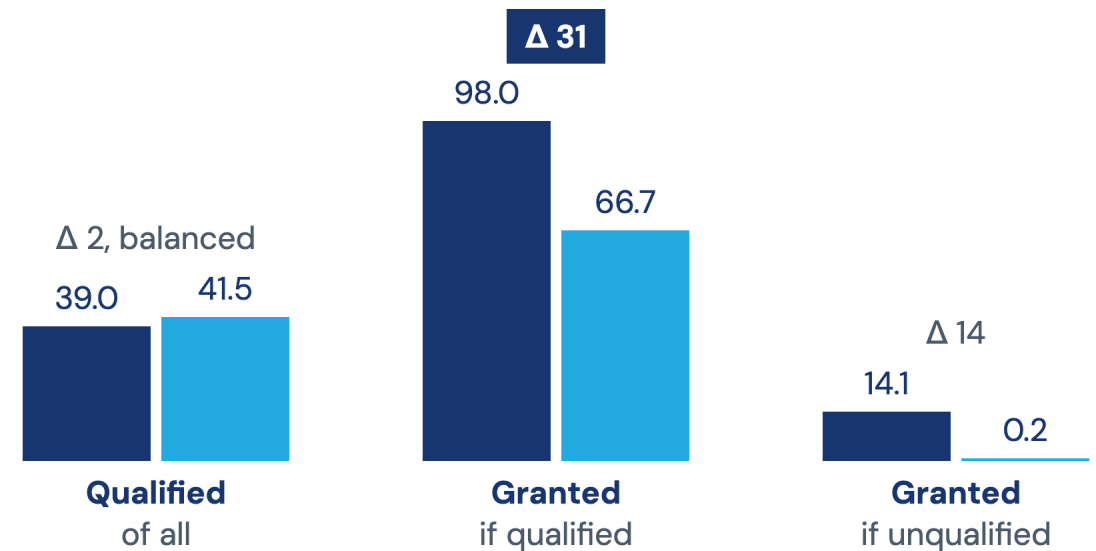
Qualified = top 40% of applicants on this score. Among qualified applicants, Centre gets far more grants.

Past committee decisions



Equal opportunity gap: 30 points

Baseline model predictions



Equal opportunity gap: 31 points

Region leaks through other columns.

We removed region_administrative and trained a Random Forest to predict Centre vs Remote from the remaining features. We tested all 127 feature combinations with test AUC.

1

What leaks alone?

Feature	AUC
Postal code	1.000
Distance	0.998
Hours worked	0.802
Income	0.669
First-gen	0.583
Program	0.557
R score	0.514

Postal code and distance reveal region on their own.

2

Is one enough?

Removed	AUC
Postal code	0.9983
Distance	1.0000
Hours worked	0.9998
Income	0.9999
First-gen	0.9998
Program	0.9999
R score	0.9999

All features: 0.9998. Removing any one changes nothing.

3

How much must go?

Removed	AUC
Postal + distance	0.841
+ income	0.811
+ hours	0.597

Without the obvious two, region is still 0.84.

4

What stays safe?

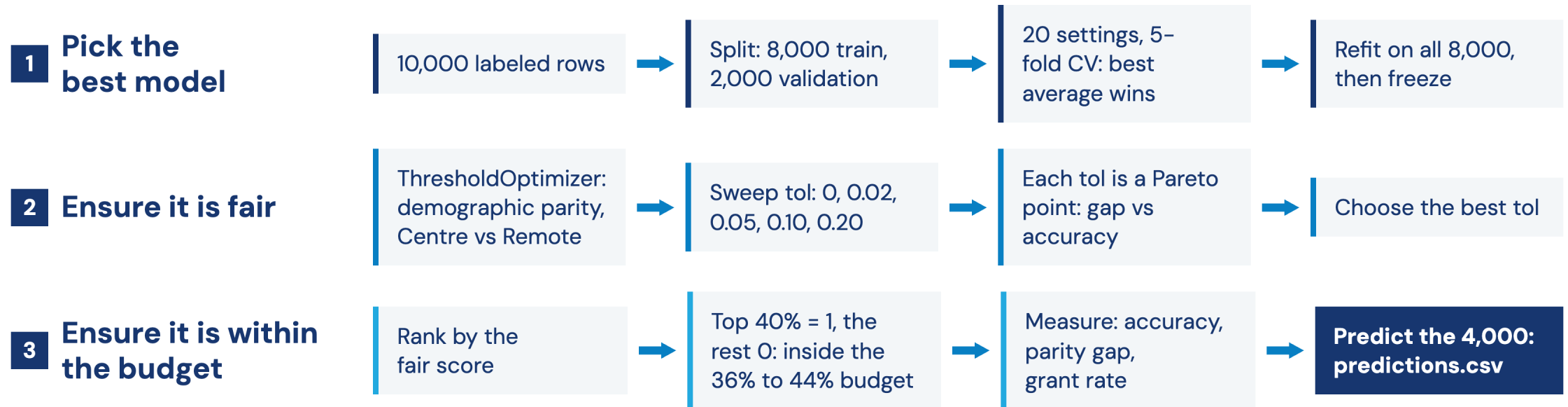
Feature set	AUC
R + first-gen + program	0.597
R + first-gen	0.572
R + program	0.539
R score only	0.514

The R score is almost region-blind.

Source: proxy_check.ipynb. Random Forest predicting Centre vs Remote, test AUC on a 30% split.

Pick it. Make it fair. Keep it in budget.

- 3 models: logistic regression, random forest, XGBoost. Main feature sets: 3 to 6.
- Each model followed this architecture.



The fairness step needs each applicant's region at decision time. Tuning uses the 8,000; the fairness sweep uses the 2,000.

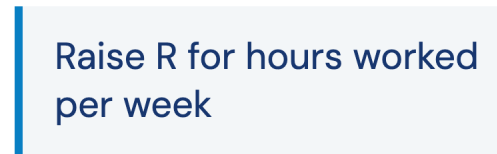
Two other ways to fix it.

2 Family 2



Why. A model trained on one region cannot learn the committee's penalty on the other.

3 Family 3



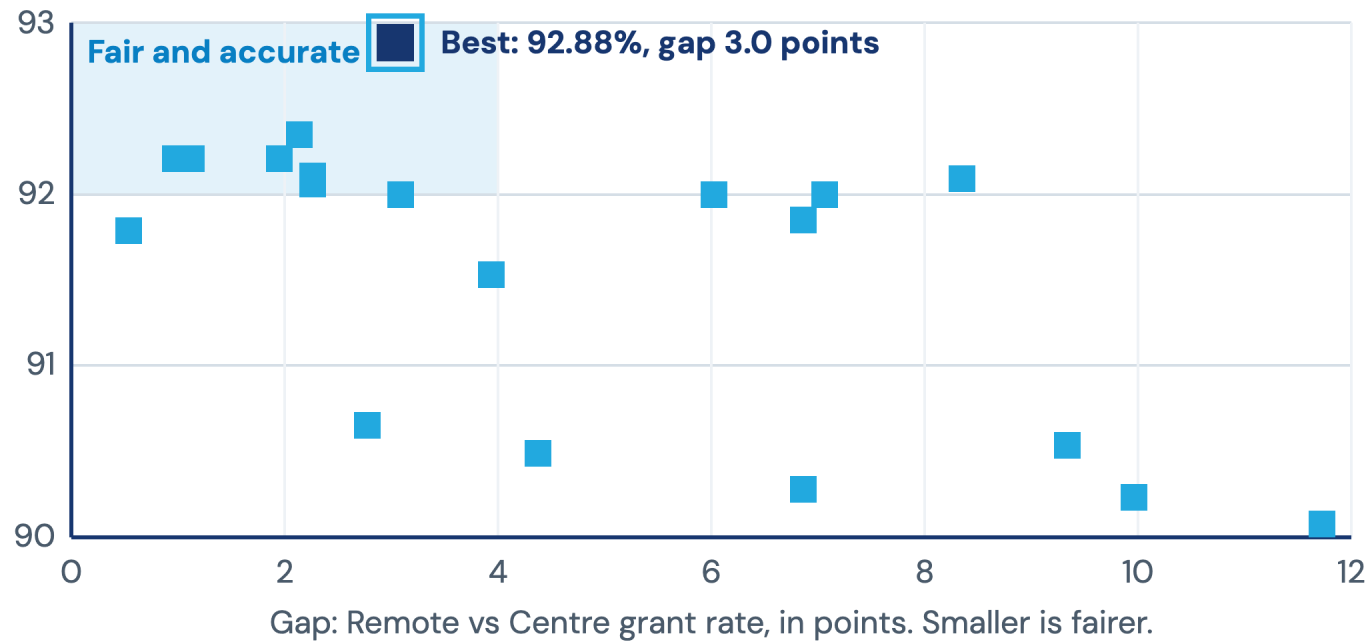
Why. Working hours affect grades, so a student who works more gets more credit on R.

■ RESULTS · HOLD-OUT SCORE AND GAP

Best score, small gap.

Each square is one run. Higher means a better F1 macro, further left means a fairer selection rate.

F1 macro (%). Higher is better.



BEST PERFORMER

92.88%

F1 macro

3 pts

Remote 39.8% - Centre 36.8%
= 3.points

**38% budget, ranked by the
qualified score**

F1 macro: HxBuddy score on the hidden hold-out, not the official points. Gap: Remote minus Centre selection rate on our 4,000 predictions, absolute.

Keep watching.

1 Track

Every month, by region

- Selection gap (total)
- Equal-opportunity gap
- Grant rate by R-score band

2 Alert

When to raise a flag

- Selection or equal-opportunity gap above 5 points
- Gap in any R band above 5 points
- Grant rate outside 36% to 44%

3 Act

Who does what

- A named owner reviews each alert
- Humans re-check Remote denials
- Specialists help improve the metric quarterly (fairness metric)

Disclosure: decisions use each applicant's region, income and hours. We say so openly.

Merci.



ÉquiAlgo: fair student financing

Code, notebooks and predictions



[github.com/NawarTurk/
ivado-student-finance-fairness](https://github.com/NawarTurk/ivado-student-finance-fairness)



Scan for the repo